

25.1 Astronomy

Question Paper

Course	CIE A Level Physics
Section	25. Astronomy & Cosmology
Topic	25.1 Astronomy
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

State what is meant by a standard candle.

[1 mark]

Question 1b

Explain how a standard candle can be used to measure astronomical distances.

[2 marks]

Question 1c

A standard candle in a distant galaxy is found to have a luminosity of $2.0 \times 10^{36} \text{ W}$ and a radiant flux intensity on Earth of $2.59 \times 10^{-15} \text{ W m}^{-2}$.

Show that the distance of the galaxy from the Earth is about $8 \times 10^{24} \text{ m}$.

[3 marks]

Question 1d

The Stefan-Boltzmann law states that the power output of a star depends on two factors.

Place a tick (✓) next to the **two** correct factors.

Mass	
Radius	
Surface temperature	
Core temperature	

[2 marks]

Question 2a

State Wien's displacement law.

[1 mark]

Question 2b

The colour of a star indicates its surface temperature. In a sample of stars, the main colours observed are

white blue orange yellow red
blue-white yellow-white

List the colours from the coolest to the hottest.

[3 marks]

Question 2c

Fig. 1.1 shows how the rate of emission for three stars, A, B and C varies with wavelength.

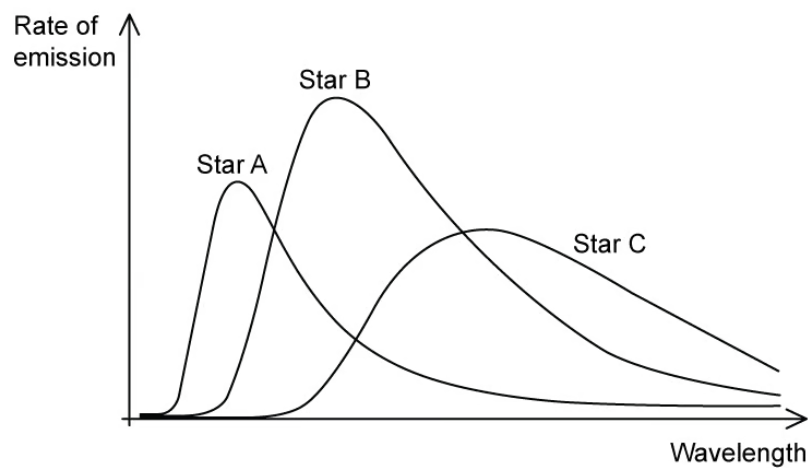


Fig. 1.1

Using Fig. 1.1, identify

(i)
the hottest star,

[1]

(ii)
the coolest star,

[1]

(iii)
the brightest star.

[1]

[3 marks]

Question 2d

Fig. 1.2 shows the variation of wavelength with intensity for four different stars.

On Fig. 1.2, identify the visible, ultraviolet and infrared regions of the graph.

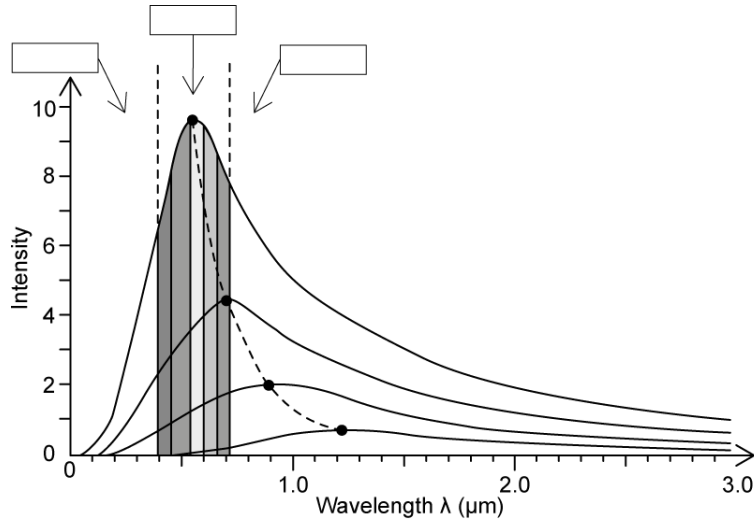


Fig. 1.2

[2 marks]

Question 3a

State what is meant by the luminosity of a star.

[1 mark]

Question 3b

The Sun emits radiation with a peak emission occurring at a wavelength of 5.2×10^{-7} m.

Show that the Sun has a surface temperature of about 6000 K.

Wien's displacement constant = 2.898×10^{-3} m K

[2 marks]

Question 3c

The radiation received from the Sun at the top of the atmosphere is 1.37 kW m^{-2} .

Show that the Sun's luminosity is about $4 \times 10^{26} \text{ W}$.

Distance from the Sun to the Earth = $1.49 \times 10^{11} \text{ m}$

[2 marks]

Question 3d

Use the Stefan-Boltzmann law to determine the radius of the Sun.

[2 marks]



Model Answers: Easy

1a

(a) A standard candle is:

- A (stellar) object of known luminosity; [1 mark]

[Total: 1 mark]

1b

(b) A standard candle can be used to measure astronomical distances by:

EITHER

- Measuring the standard candle's flux / brightness / intensity on Earth; [1 mark]
- Using the inverse square law (of flux) to calculate distance; [1 mark]

OR

- Using $F = \frac{L}{4\pi d^2}$; [1 mark]
- Where F = flux / brightness / intensity (of standard candle), L = luminosity, d = distance (to standard candle); [1 mark]

[Total: 2 marks]

1c

(c) Show that the distance of the galaxy from the Earth is about 8×10^{24} m:

List the known quantities:

- Radiant flux on Earth, $F = 2.59 \times 10^{-15} \text{ W m}^{-2}$
- Luminosity, $L = 2.0 \times 10^{36} \text{ W}$

Calculate the distance to the galaxy using the inverse square law:

- Inverse square law of flux: $F = \frac{L}{4\pi d^2}$ [1 mark]
- $d = \sqrt{\frac{L}{4\pi F}}$
- $d = \sqrt{\frac{2.0 \times 10^{36}}{4\pi \times (2.59 \times 10^{-15})}}$ [1 mark]
- $d = 7.84 \times 10^{24} \text{ m}$ [1 mark]

[Total: 3 marks]

1d

(d) The factors that affect the power output of a star are:

Mass	
Radius	✓
Surface temperature	✓
Core temperature	

- Radius; [1 mark]
- Surface temperature; [1 mark]

[Total: 2 marks]

Make sure you understand that the power output of a star depends on its radius and its surface temperature **only**

This is the Stefan-Boltzmann law, which luckily, is included on the datasheet for your exam

$$L = 4\pi r^2 \sigma T^4$$

2a

(a) Wien's displacement law states:

- The wavelength of maximum intensity (emitted by a star) is inversely proportional to its (thermodynamic) temperature

OR

- $\lambda_{max} \propto \frac{1}{T}$, where λ_{max} = wavelength of maximum intensity and T = (thermodynamic) temperature; [1 mark]

[Total: 1 mark]

Make sure not just to state the equation, but **explain** each of the variables for full marks.

2b

(b) The colours of stars from coolest to hottest are as follows:

	colour
coolest	red
	orange
	yellow
	yellow-white
	white
	blue-white
hottest	blue

- Identifies coolest = red; [1 mark]
- Identifies hottest = blue; [1 mark]
- The rest in the correct order; [1 mark]

[Total: 3 marks]

2c

(c)

(i) The hottest star is:

- Star A; [1 mark]

(ii) The coolest star is:

- Star C; [1 mark]

(iii) The brightest star is:

- Star B; [1 mark]

[Total: 3 marks]

Make sure you understand Wien's displacement law and how to apply it

$$\lambda_{max} \propto \frac{1}{T}$$

Wien's displacement law tells us:

- The **shorter** (smaller) the peak wavelength, the **higher** the surface temperature
- The **longer** (bigger) the peak wavelength, the **lower** the surface temperature

Make sure you also understand the significance of the **height** of the peak:

- The **higher** the rate of emission, the **brighter** the star
- The **lower** the rate of emission, the **dimmer** the star

2d

(d) The correct regions of the graph are as follows:

Wien's displacement law tells us:

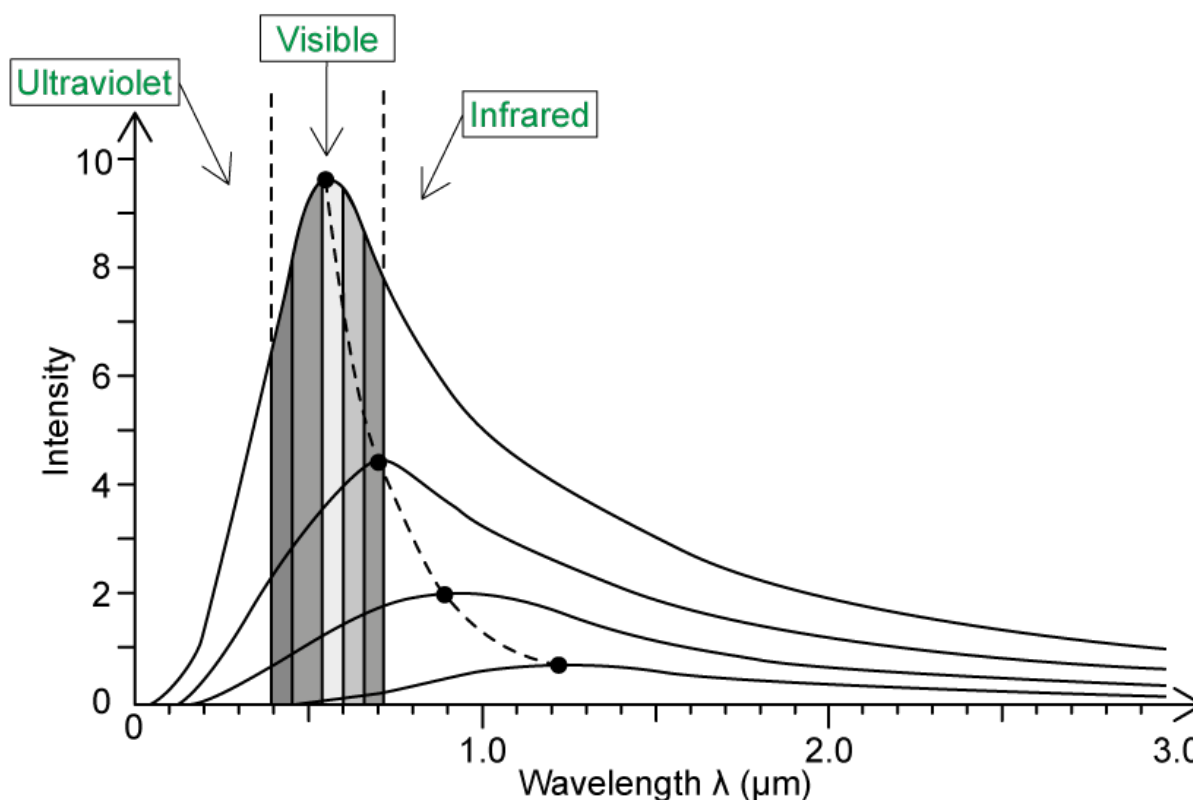
- The **shorter** (smaller) the peak wavelength, the **higher** the surface temperature
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Make sure you also understand the significance of the **height** of the peak:

- The **higher** the rate of emission, the **brighter** the star
- The **lower** the rate of emission, the **dimmer** the star

2d

(d) The correct regions of the graph are as follows:



- Visible light correctly labelled; [1 mark]
- Ultraviolet (on the left) and infrared (on the right) are correctly labelled; [1 mark]

[Total: 2 marks]

It is really important to be familiar with the relative magnitudes of wavelengths of EM radiation. It is essential that you know that ultraviolet has a shorter wavelength than visible light and infrared has a longer wavelength.

Revisit the electromagnetic spectrum topic from GCSE if you are unsure.

3a

(a) The luminosity of a star is:

- The total power of radiation emitted (by the star); [1 mark]

[Total: 1 mark]

3b

(b) Show that the Sun has a surface temperature of about 6000 K:

List the known quantities:

- Peak wavelength, $\lambda_{max} = 5.2 \times 10^{-7} \text{ m}$
- Wien's displacement constant = $2.898 \times 10^{-3} \text{ m K}$

Calculate the surface temperature using Wien's displacement law:

- Wien's displacement law: $\lambda_{max} T = 2.898 \times 10^{-3} \text{ m K}$ [1 mark]

- $$T = \frac{2.898 \times 10^{-3}}{\lambda_{max}}$$

- $$T = \frac{2.898 \times 10^{-3}}{5.2 \times 10^{-7}} = 5570 \text{ K}$$
 [1 mark]

[Total: 2 marks]

3c

(c) Show that the Sun's luminosity is about $4 \times 10^{26} \text{ W}$:

List the known quantities:

- Radiant flux intensity, $F = 1.37 \text{ kW m}^{-2} = 1370 \text{ W m}^{-2}$
- Distance, $d = 1.49 \times 10^{11} \text{ m}$

Calculate the luminosity using the inverse square law of flux:

- Inverse square law of flux: $F = \frac{L}{4\pi d^2}$ [1 mark]
- $L = F \times 4\pi d^2$
- $L = 1370 \times 4\pi \times (1.49 \times 10^{11})^2 = 3.82 \times 10^{26} \text{ W}$ [1 mark]

[Total: 2 marks]

3d

(d) Determine the radius of the Sun:

List the known quantities:

- Luminosity, $L = 3.82 \times 10^{26} \text{ W}$ (from part c)
- Surface temperature, $T = 5570 \text{ K}$ (from part b)
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Calculate the radius using the Stefan-Boltzmann law:

- Stefan-Boltzmann law: $L = 4\pi r^2 \sigma T^4$
- $r = \sqrt{\frac{L}{4\pi \sigma T^4}}$
- $r = \sqrt{\frac{3.82 \times 10^{26}}{4\pi \times (5.67 \times 10^{-8}) \times (5570)^4}}$ [1 mark]
- $r = 7.46 \times 10^8 \text{ m}$ [1 mark]

[Total: 2 marks]

